

Shelter Cluster Nepal: Monsoon preparedness assessment

Preliminary findings, Kathmandu, Nepal

**10 June
2016**



Global Shelter Cluster
ShelterCluster.org
Coordinating Humanitarian Shelter

Shelter Cluster Nepal: Monsoon preparedness assessment

- **Overview**
- **Mapping and caseload estimates**
- **Primary data collection findings**

Shelter Cluster Nepal Monsoon preparedness assessment: Preliminary findings

OVERVIEW



Context

- **Shelter Cluster Nepal is preparing** for potential flooding, landslides and displacement as a result from **monsoon rains**.
- To **inform contingency plans** for stock positioning, community mobilization, key messaging, and technical guidance > **an assessment of high-risk locations** was conducted.



What this assessment could **not** do

We could not survey a representative (random) sample of households in every VDC, here is why –

- ✓ EQ priority districts = **744** VDC/Municipalities
 - ✓ 744 VDC/Municipalities x 1 interviews = **74,400** interviews
 - Estimated interviews per day = 10
 - Duration = 14 days
- This would require:
- **Enumerators = 531**
 - **Cars = 133**

... And that is not including the 22 Terai districts...

To compare, the methodology we used required:

- **Enumerators = 12**
- **Cars needed = 5**



Assessment objectives

SECONDARY DATA

- Identify and map districts and where possible, VDCs, that are **vulnerable to monsoon outcomes** (flooding, landslides etc) and **impacts** (displacement, destruction of shelters etc) both due to previous monsoon trends and EQ impacts
- Identify **potential caseloads** due to monsoon outcomes and impacts, by district and where possible, VDC, particularly households that are living in temporary shelters following the 2015 earthquakes

PRIMARY DATA

- Understand **expectations of families** in terms of monsoon assistance from the government
- Understand level of **preparation** amongst families and options for temporary displacement
- Identify available **building materials** and respective local cost



Assessment implementation

Area: Focusing on 36 districts: 22 in Terai region, 14 EQ priority districts

SECONDARY DATA

- **Phase I (11 – 22 May):** Identify VDCs/municipalities with high monsoon vulnerability through **mapping of secondary data at lowest administrative level possible**
 - Past monsoon impacts
 - Landslide susceptibility following the 2015 EQ
 - Major rivers
 - Current food security status
 - Population remaining in vulnerable shelter conditions
- ***Calculation of caseload figures following bottom up and top down approaches***



Assessment implementation

PRIMARY DATA

- **Phase II (23 May – 1 June):** Asked shelter cluster partners to nominate case study locations that had characteristics:
 - **Part of population currently living in emergency or temporary shelters** and/or in shelters still partially damaged by natural hazard/disaster impact.
 - **Previously affected by or at risk of future landslides and/or floods.**
 - **Located in 14 earthquake affected or in the 22 Terai districts.**



In **12 locations**, the following complete:

- ✓ 1 x FGD/participatory mapping with **women living in temporary/damaged shelter** with participatory mapping
- ✓ 1 x FGD/participatory mapping with **men living in temporary/damaged shelter** with participatory mapping
- ✓ 1 x FGD/participatory mapping with **women living in permanent shelter** with participatory mapping
- ✓ 1 x FGD/participatory mapping with **men living in permanent shelter** with participatory mapping
- ✓ 1 x KI with **local authority/shelter agency**
- ✓ 1 x KI with **builder or trader in construction materials**

Assessment implementation

	FGD		KI		TOTAL
	Female	Male	LA/SC	Market	
TOTAL	26	23	12	12	73
EARTHQUAKE DISTRICTS	18	15	8	8	49
Bhaktapur	4	2	1	1	8
Bhaktapur	4	2	1	1	8
Dhading	2	2	1	1	6
Semjong	2	2	1	1	6
Dolakha	2	2	1	1	6
Laduk	2	2	1	1	6
Gorkha	2	2	1	1	6
Jaubari	2	2	1	1	6
Nuwakot	2	2	1	1	6
Rautbesi	2	2	1	1	6
Okhaldunga	2	1	1	1	5
Mamkha	2	1	1	1	5
Rasuwa	2	2	1	1	6
Ramche	2	2	1	1	6
Sindhupalchok	2	2	1	1	6
Gati	2	2	1	1	6



Assessment implementation

	FGD		KI		TOTAL
	Female	Male	LA/SC	Market	
TERAI DISTRICTS	8	8	4	4	24
Bardiya	2	2	1	1	6
Gulariya	2	2	1	1	6
Dang	2	2	1	1	6
Gadhawa	2	2	1	1	6
Mahottari	2	2	1	1	6
Gaushala	1	1			2
Jaleswor	1	1	1	1	4
Sunsari	2	2	1	1	6
Inaruwa				1	1
Paschimkasuha	2	2			4
West Kasuwa			1		1

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MAPPING AND CASELOADS ESTIMATES



Mapping and caseload outputs

VDC LEVEL

1. Hazard & Risk maps

- 22 Terai districts
- 14 EQ districts

2. Hazard, Risk & Vulnerability maps

- 22 Terai districts
- 14 EQ districts

DISTRICT LEVEL

3. Estimated caseload bottom up approach

- 22 Terai districts
- 14 EQ districts

4. Estimated caseload top down approach

- 22 Terai districts
- 14 EQ districts

Humanitarian Profile Support Guidance:

Includes methods, definitions, good practice and recommendations for establishing humanitarian population figures

April 2016

Humanitarian Population Figures

IASC Information Management Working Group

Chapter I: Notions and Definitions

- I. Importance of establishing population figures 2
- II. Current practice 2
- III. Terminology + definitions 4
- IV. Roles and responsibilities 6

Chapter II: How to estimate population figures

- I. Total Population 7
 - Sex + Age Disaggregated Data 8
- II. Affected Population 9
 - Top down + bottom up 10
- III. People in Need 11
 - The need concept 12
 - Top down + good practice 12
 - Bottom-up + aggregation rules 15

Annexes

- I. Documenting data + standards
- II. Governance principles + TORs
- III. Planning + monitoring figures
- IV. Timeline consultation process

The objective of this guidance is to provide an overview of definitions, methods and good practice on how to derive overall, inter-sectoral humanitarian population figures. These include estimates of the number of people affected by a given emergency as well as in need of, targeted by and reached with humanitarian assistance. Currently there is no collectively agreed method to establish such figures.

The guidance is intended for colleagues who are tasked with estimating humanitarian population figures and are facilitating, or participating, in joint analysis, planning and response monitoring.

This first version of the guidance focuses on estimating total populations, affected populations and populations in need of humanitarian assistance (PiN).¹ Guidance on determining figures for planning and response monitoring (populations targeted, reached and covered) is planned for inclusion in the next version (a brief overview of current practice is featured in Annex 3).²

The guidance consists of two main chapters and four annexes. The first chapter illustrates the current challenges with determining population figures and depicts the evolution of the notions and definitions of humanitarian population figures since the 2011 IASC Humanitarian Profile Guidance.³ It includes key working definitions, examples, and a section on roles and responsibilities, laying out the basic requirements for estimating humanitarian aggregate population figures. The second chapter is dedicated to concrete examples from the field and practical recommendations reflecting current best practice, approaches and methodologies.

¹ In the following abbreviated as "people in need" and PIN

² In addition to this note, further practice can be found on the IMWG's Sub-Groups' space on HR.info <https://www.humanitarianresponse.info/en/topics/imwg/imwg-sub-groups>

³ See IASC Guidance on the Humanitarian Profile Common Operational Data Set, June 2011: https://www.humanitarianresponse.info/en/system/files/documents/files/iasc_guidelines_on_the_humanitarian_profile_common_operational_dataset_2012-08-07_EN.pdf

A) TOP DOWN: In the immediate aftermath of a shock such as an earthquake, sub-groups may not yet be determined and figures for population displaced and casualties may not be available. In these cases, the population affected is determined as a total, then disaggregated as data becomes available.

Using location and crisis impact to estimate people affected	Location	Pre-crisis vulnerability	Crisis Impact	Intensity of needs
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For the 2015 Myanmar Humanitarian Needs Overview, the affected population was calculated as total population living within flood affected areas, looking at areas of probable standing flood waters. Additionally, using spatial analysis (intersection) the number of flood affected villages and roads could be also calculated.

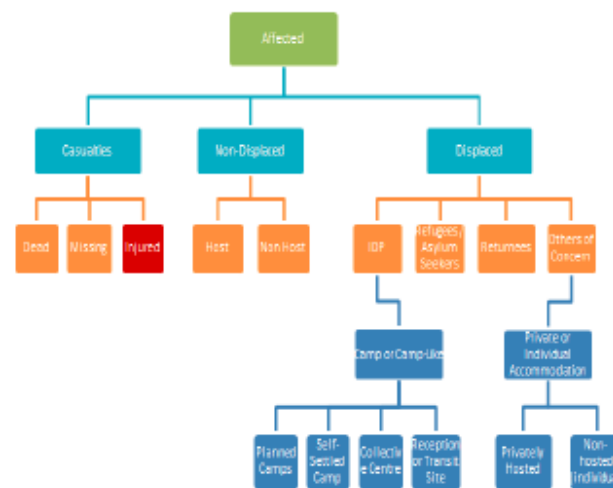
Spatial analysis is increasingly used when estimating the size of populations affected by conflict. In the 2015 Libya Humanitarian Needs Overview (HNO), the number of affected people was calculated by extracting the number of people living within 2 kilometres of a conflict incident between June 2014 and 2015, as recorded in a regional conflict incident database (ACLED). In addition, 80% of the population of specific provinces was added, where all medical facilities were reported to be non-functional and thus unavailable to attenuate conflict effects such as conflict-related injuries. The total affected population estimate further includes IDPs, refugees and asylum seekers and migrants impacted by the conflict.

This example of the humanitarian profile created through a top-down approach is recommended where little in-crisis information, time and resources are available and disaggregation per population group is not yet possible. The disaggregation and thus mitigation of overlap in the above example is done based on a spatial basis or being part of one of the population groups affected by the conflict. However, when both analytical elements are combined, the risk of overlap increases: an IDP residing within the 2 km of a conflict incident, for example, will be double counted as affected.

B) BOTTOM UP:

Figure 5: Humanitarian Profile (IASC IMWG 2011)

The Humanitarian Profile framework (Figure 5) outlines the components of estimating affected population. The number of affected people is defined as the sum of casualties (excluding the category injured, as it is not mutually exclusive), displaced population and non-displaced affected population. The sub-groups of population affected are not static, but determined specifically in the crisis context and thus will vary. Once these mutually exclusive groups are identified and figures sourced for each group, their sum results in the overall number of People Affected. The non-affected population is derived by subtracting the number of people affected from the total population.



Mapping and caseload estimates

1. Hazard & Risk maps

TERAI DISTRICTS

■ METHODOLOGY

1. Map VDCs where people are at risk of flooding according to **DPRP risk** tables
2. Map VDCs according to number of years since 2000 when **flood events affected people** according to NRCS/MOHA/DesInventar datasets
3. Compare DPRP categorisation and events data: identify **VDC risk category** as the highest between the two
4. Triangulate with **Rivers** and **ICIMOD Flood modelling** to check that no VDC with inundation risk is categorised as low





Flood risk in VDCs within Terai region



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REACH

AN INITIATIVE OF
IMPACT INITIATIVES
AFTER THE DISASTER

Terai Flood Risk

- High
- Medium
- Low
- No Risk
- Reserve and Park
- District boundary

0 15 30 60 KM

Mapping and caseload estimates

1. Hazard & Risk maps

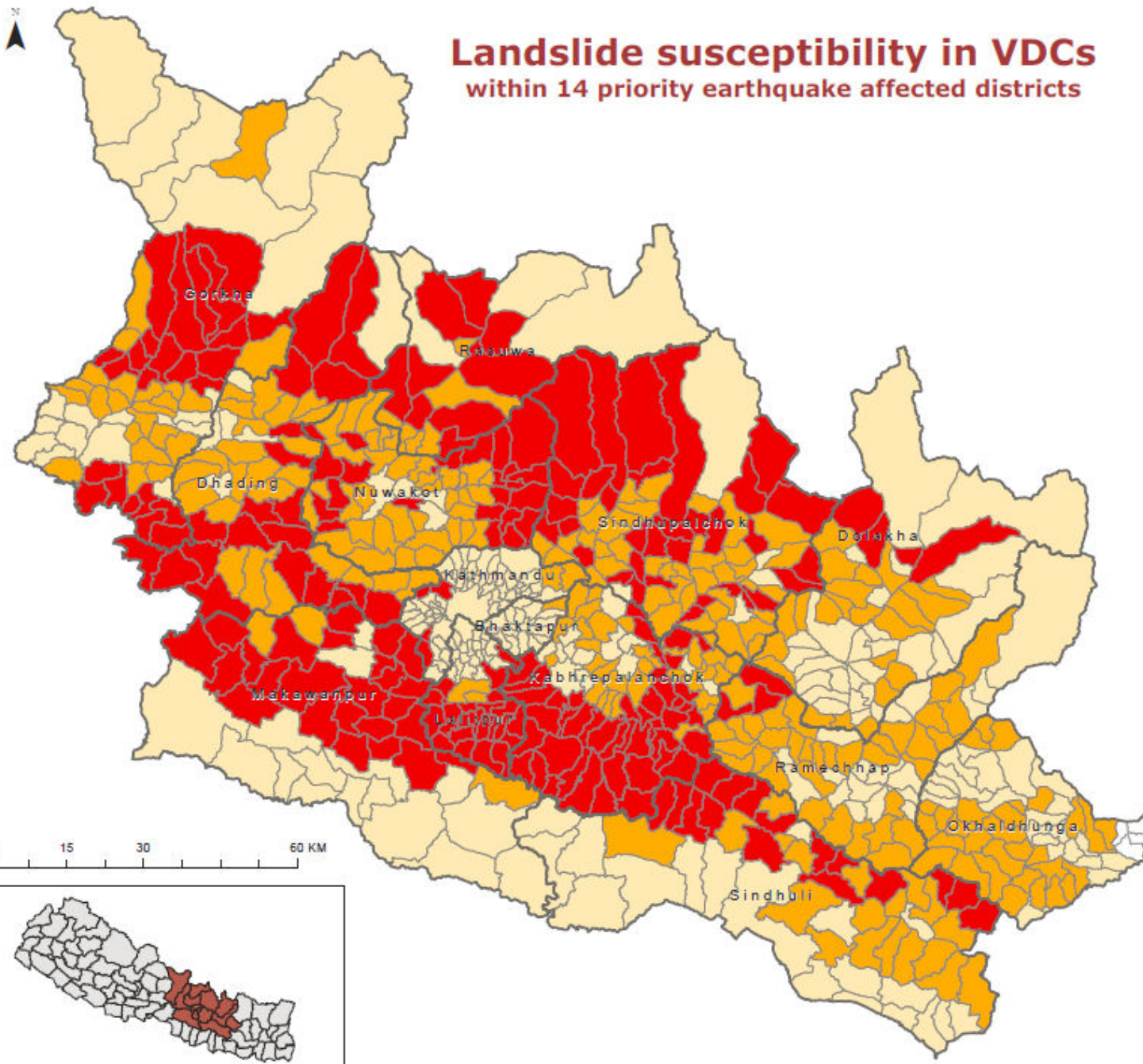
EARTHQUAKE DISTRICTS

■ METHODOLOGY

1. Map **ICIMOD landslide susceptibility data** set showing majority % of areas within ICIMOD hazard category high, moderate low
2. Compare susceptibility data with **other hazard data** (**GoN/Durham / Michigan University**) to make sure no areas are missed
3. Compare with impact data to produce risk map (confirming human impact) – **DesInventar/NRCS** (being cleaned); **Durham University** (awaiting)



Landslide susceptibility in VDCs within 14 priority earthquake affected districts



Landslide susceptibility

- High probability
- Moderate probability
- Low probability
- No data
- District boundary

Data sources:
Coordinate System: WGS 1984 UTM zone 45N
Projection: Transverse Mercator
Datum: WGS 1984

Developed by:

Mapping and caseload estimates

2. Hazard, Risk & Vulnerability maps

TERAI & EARTHQUAKE DISTRICTS

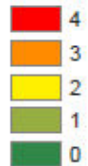
■ METHODOLOGY

1. Use **Hazard & Risk maps** as basis: Low risk (1), Moderate risk (2), High risk (3)
2. Add **IPC food security data**, increasing score by:
 - 1 level for moderate food insecurity and
 - 2 levels for severe food insecurity (low or no food insecurity we do not adjust for)
3. Add location of IDP camps – *to be done*



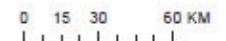
Terai region: Flood risk and food insecurity

Vulnerability Score

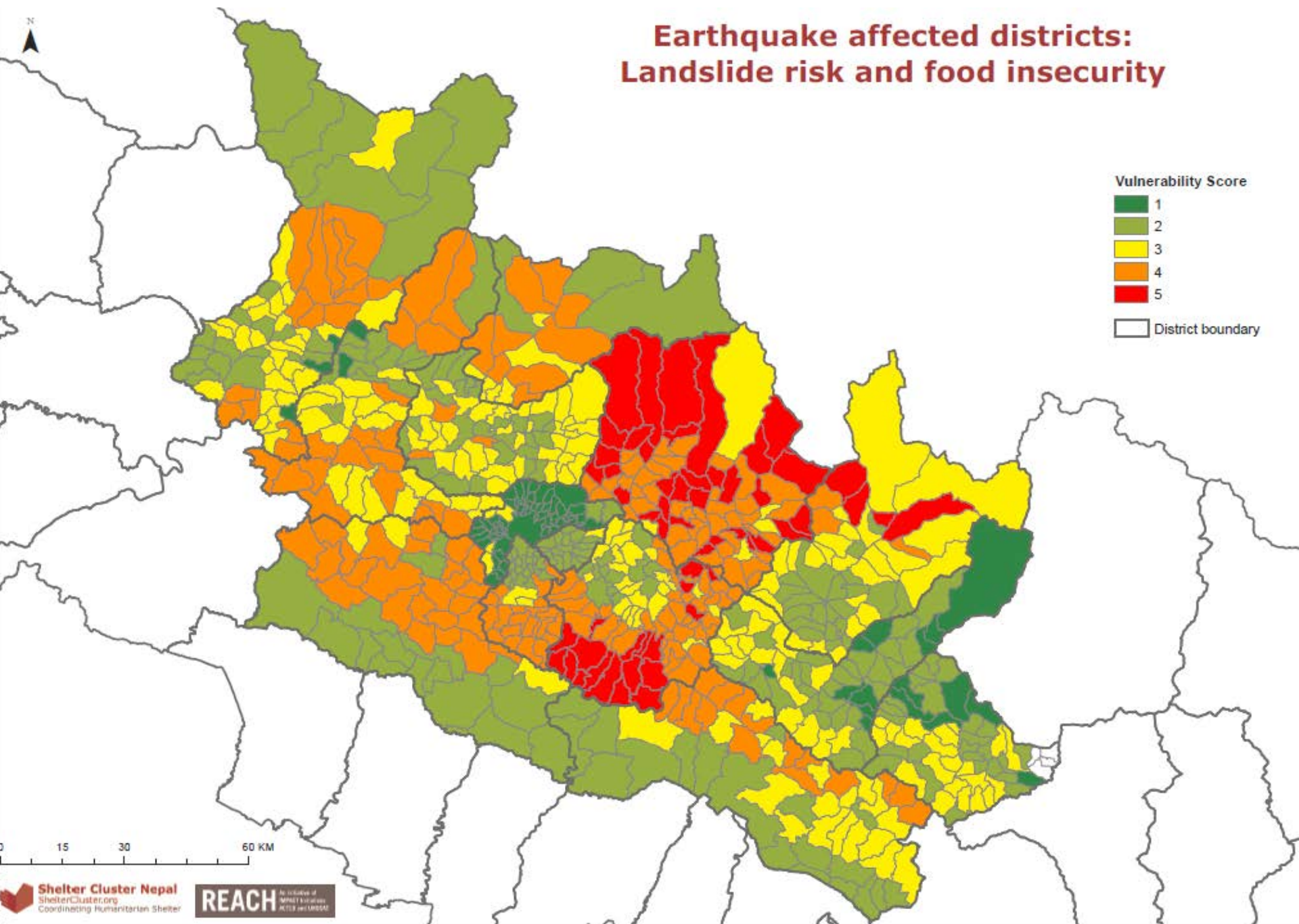


 Reserve and Park

 District boundary



Earthquake affected districts: Landslide risk and food insecurity



Mapping and caseload estimates

3. Estimated caseload: **bottom up** approach

- *STATUS: First draft produced (see next slide); **awaiting full access to DesInventar historical data***

TERAI DISTRICTS

■ METHODOLOGY

1. Compare **DesInventar**, **NRCS & MoHA events data**, identify;
 1. The data source with the **highest number of recorded events** for each year;
 2. From the highest numbers recorded for each year, identify **the lowest number** of households with fully and partially destroyed homes across recorded years in each district
 - *Convert to individuals by assuming on average 5 individuals per household*
 - ***Adjust in line with population growth rate***
 3. Repeat for the year with the **highest number** of households with fully and partially destroyed homes
 - *Convert to individuals by assuming on average 5 individuals per household*
 - ***Adjust in line with population growth rate***



Estimated caseload: **bottom up** approach - Terai

Year	NRCS				MOHA RCO				MAX
	Districts	Houses destroyed	Houses damaged	Total	Districts	Houses destroyed	Houses damaged	Total	
2000	47	3719	1670	5389					5389
2001	3								
2002	20	4634		4634					4634
2003	32	1184	2847	431					4031
2004	14	21678	43339	65017					65017
2005	18	481	735	1216					1216
2006	13		761	761	49	910	8098	9008	9008
2007	35	2258	40951	639	125	8693	1120	9813	63009
2008	19	916	591	157	282	12950	1643	14593	14593
2009	26	1363	2791	4154	184	413	3497	3910	4154
2010	162	1270	2058	3328	276	2406	5727	8133	8133
2011	31	1377	2047	3424	240	2699	3817	6516	6516
2012	23	338	1305	1643	47	170	309	479	1643
2013	12	623	2004	2627	266	263	108	371	2627
2014	11	9337	25225	34562	65	8622	24447	33069	34562



Estimated caseload: **bottom up** approach - Terai

Records 2000 - 2014	Average homes destroyed per affected district	Average homes damaged per affected district	Total damaged / destroyed homes per affected district	Total damaged / destroyed homes	Estimated individuals with damaged/ destroyed housing (assuming 5 HH size)
Max - 2004	1548	3096	4644	65017	325,085
Average (of highest records 2000-2014)	181	302	483	15016	75,079
Min - 2005	27	41	68	1216	6,080

*****To be calculated for each individual district**



Mapping and caseload estimates

3. Estimated caseload: **bottom up** approach

- *STATUS: **awaiting Durham / NRCS historical data***

EARTHQUAKE DISTRICTS –

■ METHODOLOGY

1. Using **NRCS** or **Durham historical impact** data (recorded data before AND after EQ) identify
 1. The year with the **lowest number** of recorded landslide events before EQ
 2. Compared to that year, identify the **% increase in number of landslides** causing human impacts in each district in 2015
 3. Compared to that year, predict the number of affected households by **increasing the number of affected households with the same % increase as seen in the number of landslides** – applied on the highest recorded figures across DesInventar, NRCS and MoHA data sets (same as Terai methodology, see previous slide)
2. Repeat for the year with the **highest number** of recorded landslide events

Mapping and caseload estimates

4. Estimated caseload top down approach

- *STATUS: First draft produced (see next slide); add population living at risk of flooding (awaiting ICIMID or Michigan flood risk data)*

EARTHQUAKE DISTRICTS

- **METHODOLOGY**
- Use District level Department of Civil Registration or World Bank data to identify base case population 'zooming in' on vulnerable individuals
 - *We have also used total census figures for comparison— see next slides*
- Apply % living in temporary shelters as of Shelter Cluster / REACH September 2015 assessment
- Use the Hazard/Risk mapping and projected census data to identify the % of district population that live in VDCs identified as high/moderate/low risk of landslides **(and flooding, pending data)**
- Apply the % of district population to the number of vulnerable individuals in each District

Estimated caseload: **top down** approach – 14 EQ Districts

The most vulnerable population (according to social protection) living in temporary shelters and collective centers in areas with High/Medium/Low hazard of landslides

District	High risk	Moderate risk	Low risk
Bhaktapur	0	92	5963
Dhading	10236	11770	1784
Dolakha	732	6009	8677
Gorkha	9036	9298	11073
Kathmandu	72	22	3567
Kavrepalanchok	6960	4788	6579
Lalitpur	276	60	2259
Makawanpur	829	111	1943
Nuwakot	6392	14604	4162
Okhaldhunga	0	2199	1703
Ramechhap	1119	11316	6872
Rasuwa	1445	1592	199
Sindhuli	904	1463	2667
Sindhupalchok	10677	15334	2046
TOTAL	48,679	78,659	59,494



Estimated caseload: **top down** approach – 14 EQ Districts

TOTAL Population living in temporary shelters and collective centers in areas with High/Medium/Low hazard of landslides			
District	Caseload High risk	Caseload Moderate risk	Caseload Low risk
Bhaktapur	0	1752	113039
Dhading	93129	107090	16229
Dolakha	7906	64878	93681
Gorkha	78126	80389	95732
Kathmandu	2410	751	119310
Kavrepalanchok	83460	57416	78896
Lalitpur	6394	1389	52289
Makawanpur	19679	2630	46113
Nuwakot	62822	143525	40900
Okhaldhunga	0	20400	15801
Ramechhap	9990	101050	61364
Rasuwa	15781	17393	2175
Sindhuli	11206	18125	33046
Sindhupalchok	102255	146859	19595
TOTAL	493,158	763,647	788,172



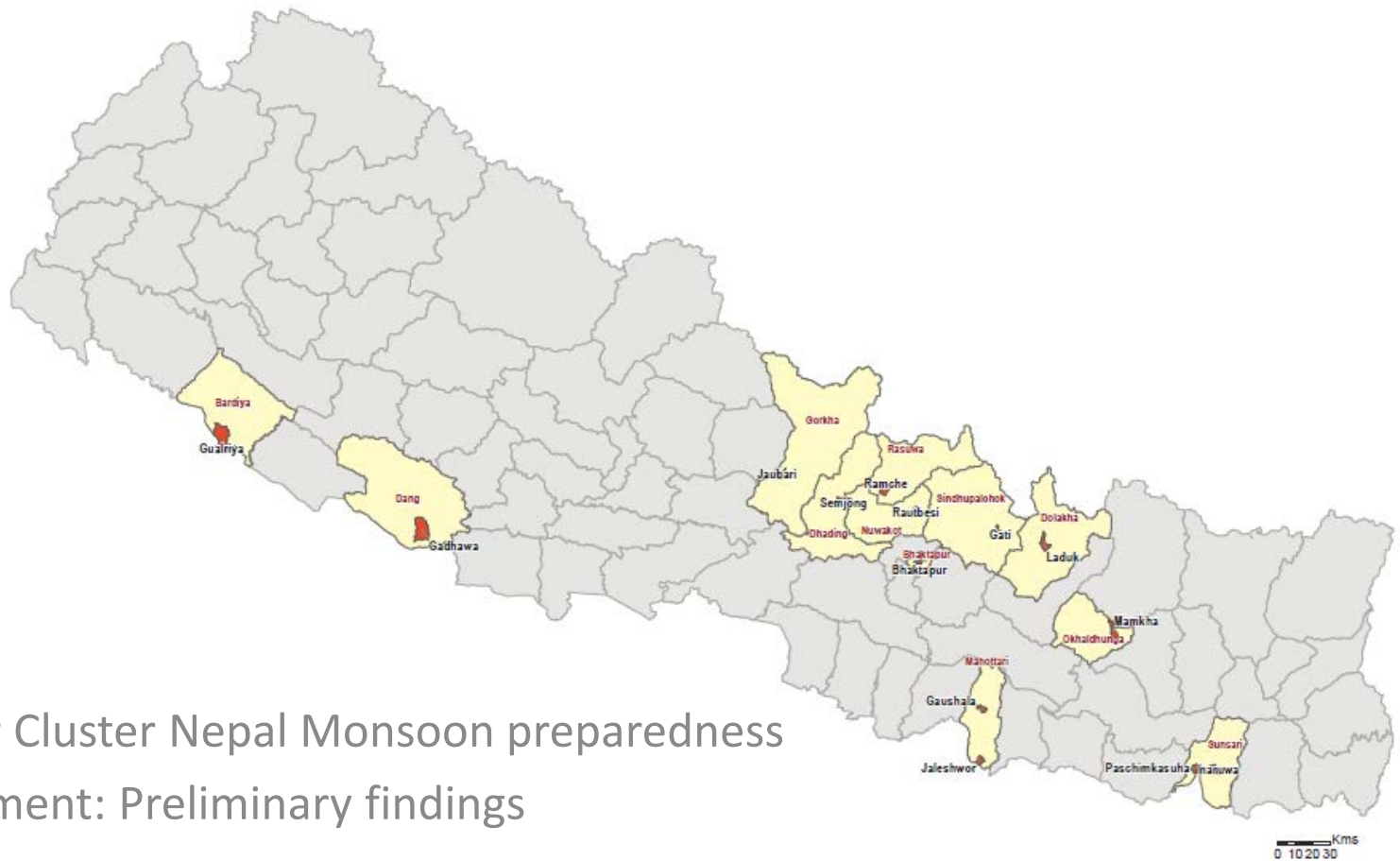
Mapping and caseload estimates

4. Estimated caseload **top down** approach

- ***STATUS: awaiting data from Department of Civil Registration, expected 13/14 June***

TERAI DISTRICTS

- **METHODOLOGY**
- Use District level **Department of Civil Registration data** to identify base case load of **vulnerable population**
- Use the **Hazard/risk mapping** and **projected census data** to identify the **% of district population** that live in VDCs identified as high/moderate risk
- Apply **the % of district population** to the **number of vulnerable individuals** in each District



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PRIMARY DATA COLLECTION FINDINGS



How to interpret the qualitative data analysis tables

Source types:

ALL = (FGD + KI)

PT = FGD in Permanent shelters

TY = FGD in Temporary shelters

		HADING			DOLAKHA			GORKHA			NUWAKOT			OKHALDU NGA			RASUWA			SINDHUPA LCHOK		
Source type																						
	ALL/TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY
PREVIOUS EVENTS																						
Flooding > FGD/KI reported past flood events here	1				1	1	1	1														
Landslide > > FGD/KI reported past flood events here		1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
PREVIOUS IMPACTS																						
crops/land damaged	1	4	2	2	2	1	1	5	2	3	5	2	2	3	1	2	6	2	3	3	1	1
houses damaged/destroyed	4	1	1		4	2	1	3	1	1	2	1	1	4	1	2	6	2	3	4	2	1
roads/pathway/bridges damaged/destroyed		5	1	3	5	2	2	3	1		5	1	2	1			5	1	3	6	2	1

'4' = number of FGDs/KI interviews where topic mentioned

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PREVIOUS EVENTS & IMPACTS



PREVIOUS EVENTS AND IMPACTS

“There was a scary lightening, and then we heard rumbling sound at 4:30 in the morning. Within 1 to 15 minutes we were already drowned in the water up to the neck.

It was hard for differently able people to escape. As we were acquainted with the place, we managed to swim to higher ground”

[FGD participant describing the flooding of temporary shelter site at Maheshwori football ground, Bhaktapur, April 2015]



EQ PRIORITY DISTRICTS	BHAKTA PUR	DHADING			DOLAKHA			GORKHA			NUWAKOT			OKHALDU NGA			RASUWA			SINDHUPA LCHOK		
Source type	ALL/TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY
PREVIOUS EVENTS																						
flooding	1				1	1	1	1	1	1	1	1	1	1		1				1	1	1
landslide		1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
PREVIOUS IMPACTS																						
crops/land damaged	1	4	2	2	2	1	1	5	2	3	5	2	2	3	1	2	6	2	3	3	1	1
houses damaged/destroyed	4	1	1		4	2	1	3	1	1	2	1	1	4	1	2	6	2	3	4	2	1
roads/pathway/bridges damaged/destroyed		5	1	3	5	2	2	3	1		5	1	2	1			5	1	3	6	2	1
water source damaged/contaminated	2	5	2	2	4	2	1	4	2	2	5	2	2	3	1	1	3	1	2	1		
school damaged/destroyed	1	1		1	2		1	2		1	2	1	1	4		3	1		1			
fatalities		3	1	1				1		1				1		1	4	1	2	1	1	
livestock swept away		2		2							3		2	2	1	1	2		2			
health facility damaged/cut-off	2							1			1									1		1
belongings washed away/spoiled/not accessible	2																1		1			
access to markets construction materials cut-off		1						1						1			1			1		
land permanently destroyed/impossible to inhabit		1	1														1	1				
electricity cut off														2	1							
temple damaged	1																					
drainage/sewage system damaged	1																					

TERAI DISTRICTS	BARDIYA			DANG			MAHOTTARI			SUNSARI		
Source type	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY
PREVIOUS EVENTS												
flooding	1	1	1	1	1	1	1	1	1	1	1	1
landslide												
PREVIOUS IMPACTS												
houses damaged/destroyed	6	2	3	4	2	1	6	3	2	5	3	1
crops/land damaged	4		3	7	3	2	2	1		4	2	1
roads/pathway/bridges damaged/destroyed	4	1	2	6	2	2	3	1		4	2	1
belongings washed away/spoiled/not accessible	3		3	2	2		3	2	1	3		2
school damaged/destroyed	3	1	2				1	1		4	1	2
water source damaged/contaminated	1	1					1	1		3	1	1
without food during initial days				1	1		1		1	2	1	1
fatalities	4		3									
soil erosion				1		1	1		1	2		1
livestock swept away	1						1		1	1		1
land permanently destroyed/impossible to inhabit	1			1	1					1	1	
health facility damaged/cut-off				1			2			1		
cooking facilities damaged							2		2			
access markets construction materials cut-off										1		
electricity cut off							1					
temple damaged										1		
phone network cut off							1					

CURRENT LIVING CONDITIONS

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EQ PRIORITY DISTRICTS	BHAKTA PUR	DHADING	DOLAKHA	GORKHA	NUWAKOT	OKHALDU NGA	RASUWA	SINDHUPA LCHOK
	ALL/TY	ALLPTTY	ALLPTTY	ALLPTTY	ALLPTTY	ALLPTTY	ALLPTTY	ALLPTTY
PREVIOUS EVENTS								
flooding	1		111	111	111	1 1		111
landslide		111	111	111	111	111	111	111
WATER								
tap - spring or main network	2	4 2 2	1 1		5 2 2	4 1 2	2 1 1	1 1
river			3 2 1	4 2 2	2 1	1		2 1 1
well/handpump	6		1 1				1 1	
water contaminated/restricted availability	2		4 1 2	1 1	1		4 2 1	3 1 1
TOILETS								
temporary/poor condition	1	1 1		3 1 2	2 1 1			1 1
shared/communal	2	2 2			2 2		1 1	
none	1						1 1	
insufficient toilets	2							
SHELTERS								
warm/dry	2	1 1	2 2	1 1	3 2 1		2 2	3 2 1
weak/leaking roof/water inside	7	3 1 2	1 1	3 2 1	2 2	3 1 2	2 2	1 1
unsafe	1	1 1	3 1 2	1 1	2 2	1 1	2 1 1	3 2 1
too hot/cold	2	2 2	2 2	3 2 1	1 1		1 1	1 1
insufficient space	1							
mosquitos	1							

[illegible]

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CURRENT DISPLACEMENT



EQ PRIORITY DISTRICTS	BHAKT APUR	DHADING			DOLAKHA			GORKHA			NUWAKOT			OKHALDU NGA			RASUWA			SINDHUPA LCHOK		
	ALL/TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY
PREVIOUS EVENTS																						
flooding	1				1	1	1	1	1	1	1	1	1	1		1				1	1	1
landslide																						
		1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
ONGOING DISPLACEMENT																						
due to landslide		2		1	1	1		5	2	2	1		1	3	1	1	11	4	4	2	1	1
due to EQ	7				3	1	2	1		1				1	1		2	1	1			
same ward	2	1		1	2		1	1		1		1								1		1
due to flooding								1			4	1	2									
due to lack of services					1	1								3	1	2						
original homes totally destroyed	4				1	1											1	1				
due to lack of land														1		1						

TERAI DISTRICTS	BARDIYA			DANG			MAHOTTARI			SUNSARI		
	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY
PREVIOUS EVENTS												
flooding	1	1	1	1	1	1	1	1	1	1	1	1
landslide												
ONGOING DISPLACEMENT												
due to flooding				2	1		4	2	1	5	3	1
due to lack of services				1		1				1		1
original homes totally destroyed	2		2									
same ward	2		2									
due to lack of food				1		1						

Common Messages for Disaster Risk Reduction in Nepal

Nepal Risk Reduction Consortium Communications Group

10 Key Messages for Floods

1. Ask your local authorities if your community has a flood early warning system. If so, understand the early warning system to help prepare for floods.
2. Are you ready? Prepare a flood evacuation plan with your family. Agree on a meeting point at a safe place on high ground outside your home, such as a community flood platform if you have one. Establish safe evacuation routes to the meeting point for your family and livestock

3. Have an emergency bag ready for your family. This should contain essential items you will

10 Key Messages for Landslide

1. Have an emergency bag ready for your family. This should contain essential items you will need immediately after a flood or landslide. Contents: a torch and batteries, a small radio, enough dry food such as beaten rice or instant noodles for one day, a plastic bottle of drinking water, a bottle of Piyush chlorine drops for purifying drinking water, a basic medical kit and photocopies of your ID cards. Store the bag in a safe place at least 3 feet above ground.
2. Carry a whistle on your person at all times to attract attention in an emergency. Blow the whistle if you are trapped or injured by an earthquake, flood or landslide. You can also blow the whistle if you are attacked. A whistle can be heard far away and it can save your voice. Keep the whistle on your key ring or hang it on a chord round your neck.
3. Mitigate your landslide risk by avoiding cutting down trees. Practicing land conservation on farmed hillsides such as terracing is more effective for growing crops.

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DRR MESSAGE AWARENESS



EQ PRIORITY DISTRICTS	BHAKT APUR	DHADING			DOLAKH A			GORKHA			NUWAKO T			OKHALD UNGA			RASUWA			SINDHUP ALCHOK		
	ALL/T Y	AL L	PT	TY	AL L	PT	TY	AL L	PT	TY	AL L	PT	TY	AL L	PT	TY	AL L	PT	TY	AL L	PT	TY
PREVIOUS EVENTS																						
flooding	1				1	1	1	1	1	1	1	1	1	1		1				1	1	1
landslide			1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
DRR MESSAGE AWARENESS																						
aware of high risk areas	5	4	2	2	4	2	2	4	2	2	4	2	2	3	1	2	4	2	2	4	2	2
know what to do if flooding/landslide	5	4	2	2	4	2	2	2	2		3	2	1	3	1	2	3	2	1	4	2	2
look out for warning signs of landslide/soil erosion risk	1	2	1	1	4	2	2	4	2	2	4	2	2	3	1	2	2	1	1	2	1	1
listen to radio to monitor weather/hear warnings	3	2		2	3	2	1				4	2	2	2	1	1	2	2		4	2	2
know what to do if landslide trapped	2	2	1	1	1	1		3	2	1	3	2	1	3	1	2	2	2		2		2
taking action to reduce landslide risk	2	2	2		2		1	5	2	2	1		1	3	1	2	4	2	1	1		
look out for warning signs of flood risk	3				4	2	2	4	2	2	4	2	2	1		1				2		2
taking action to reduce flood risk	1				3	2		5	2	2	1	1		3	1	2	1					
know who will warn/who to ask about threats														2	1	1	1	1				
have emergency bag	2																			1		1

“Roads and land are at risk of extinction here, in 2 years there will be no Mamkha VDC”

[FGD participant describing how landslides are demolishing the hill VDC where she lives, Mamkha, Okhaldunga]



TERAI DISTRICTS	BARDIYA			DANG			MAHOTTARI			SUNSARI		
	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY
PREVIOUS EVENTS												
flooding	1	1	1	1	1	1	1	1	1	1	1	1
landslide												
DRR MESSAGE AWARENESS												
know what to do if flooding/landslide	4	2	2	3	1	2	5	2	3	6	3	3
aware of high risk areas	4	2	2	3	2	1	4	2	2	3	2	1
look out for warning signs of flood risk	2		2	4	2	2	3	1	2	1		1
know who will warn/who to ask about threats	2		2	1	1		2		2	1	1	
look out for warning signs of landslide/soil erosion risk							1	1		3	1	2
listen to radio to monitor weather/hear warnings				2		2	1	1				
taking action to reduce flood risk	1	1		2	1	1						
know what to do if landslide trapped							2	1	1			
have emergency bag				1	1							
taking action to reduce landslide risk										1	1	

“As there is no any ways of being safe [from the flood] we just can climb up tree once it occurs.”

[FGD participant describing lack of safety from flooding, Gadhawa, Dang]

Shelter Cluster Nepal Monsoon preparedness assessment: Preliminary findings

FEELING SAFE

EQ PRIORITY DISTRICTS	BHAKT APUR	DHADIN G			DOLAK HA			GORKH A			NUWAK OT			OKHAL DUNGA			RASUW A			SINDHU PALCHOK		
	ALL/TY	AL L	P T	T Y	AL L	P T	T Y	AL L	P T	T Y	AL L	P T	T Y	AL L	P T	T Y	AL L	P T	T Y	AL L	P T	T Y
PREVIOUS EVENTS																						
flooding	1				1	1	1	1	1	1	1	1	1	1		1				1	1	1
landslide																						
		1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
FEELING SAFE																						
No - shelter is not strong enough	1	2		1	2		2	3	1	2	2		2				3	1	2	1		1
No - risk of landslide/death					4	3	1							3	1	2	1	1		2	1	1
No - lack of available/identifyable safe land	2										4	2	2	1		1	1		1		1	
No - risk of flooding/drowning	1				3	2	1				1		1							1		1
No - heavy rainfall	1	2	2											1	1		2		2			
Yes - compared to previous location/before								1	1													

TERAI DISTRICTS	BARDIYA			DANG			MAHOTTARI			SUNSARI		
	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY
PREVIOUS EVENTS												
flooding	1	1	1	1	1	1	1	1	1	1	1	1
landslide												
LAND TENURE												
No - risk of flooding/drowning	3	2		4	2	2	2	1	1	4	2	1
No - lack of safe evacuation route	1	1		1	1							
No - heavy rainfall	1		1							1	1	
No - lack of available/identifiable safe land										1		1
Yes - compared to previous location/before	1		1									
Yes - due to new flood barrier							1					



Shelter Cluster

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PREPAREDNESS



EQ PRIORITY DISTRICTS	BHAKTA PUR	DHADING	DOLAKH A	GORKHA	NUWAKO T	OKHALD UNGA	RASUWA	SINDHU PALCHOK
Source type	ALL/TY	AL L PT TY	AL L PT TY	AL L PT TY	AL L PT TY	AL L PT TY	AL L PT TY	AL L PT TY
PREVIOUS EVENTS								
flooding	1		1 1 1	1 1 1	1 1 1	1 1		1 1 1
landslide		1 1 1	1 1 1	1 1 1	1 1 1	1 1 1	1 1 1	1 1 1
PREPAREDNESS ACTION TAKEN NOW								
strengthening roofs/structure	3	4 1 1	3 2	3 1	1 1	2 1	4 1 1	4 1 1
storing food/stock	2	2 1			2 1 1	1		1
cover roof/tent with plastic	2					2 1 1		1
building small barriers/sandbags	1				1 1	1		
building temporary shelter in safe site						2 1		
building large barrier			1 1					
PREPAREDNESS COMPARED TO PREVIOUS YEAR								
more than last year	2			4 1 2		4 1 1	3 1 1	1
same as last year	1						1 1	
less than last year		1 1	3 1		4 1 1	1 1		2 1 1
REASONS FOR LACK OF PREPAREDNESS								
lack of money/resources	1	5 1 2	2 2	2 1	1	1	3 2 1	2 2
lack of safe site	2	2	1 1			1 1	1 1	1 1
improved access to/nearby markets leaves no need		1	2 2	2 1 1			2 1	
lack of safe storage space	1	1 1						

TERAI DISTRICTS	BARDIYA			DANG			MAHOTTARI			SUNSARI		
	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY
PREVIOUS EVENTS												
flooding	1	1	1	1	1	1	1	1	1	1	1	1
landslide												
PREPAREDNESS ACTION TAKEN NOW												
strengthening roofs/structure	5	1	2	5	2	2	1	1		1	1	
cover roof/tent with plastic							4	2	2	2	1	1
moving children/belongings to raised floors	1	1		3	2	1						
early warning system	1			2			1			1		
storing food/stock				1	1		1			1	1	
building small barriers/sandbags	1			1								
family member stays awake during rain				1								
PREPAREDNESS COMPARED TO PREVIOUS YEAR												
more than last year	2	1		1	1							
same as last year							1	1				
REASONS FOR LACK OF PREPAREDNESS												
lack of money/resources	2	1		2		1	5	3	2	3	1	2
lack of safe site												
lack of safe storage space	2		2	1	1					1		1

MONSOON NEEDS

Shelter Cluster Nepal Monsoon preparedness assessment: Preliminary findings



EQ PRIORITY DISTRICTS				BHAKT APUR			DHADING			DOLAKH A			GORKHA			NUWAKOT			OKHALDUNGA			RASUWA			SINDHUPALCHOK		
				ALL	TY		AL	PT	TY	AL	PT	TY	AL	PT	TY	AL	PT	TY	AL	PT	TY	AL	PT	TY	AL	PT	TY
PREVIOUS EVENTS																											
flooding				1						1	1	1	1	1	1	1	1	1	1		1				1	1	1
landslide							1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
MONSOON NEEDS																											
construction materials/CGI				1			4	1	2	1		1	5	2	2	3	1	2	4	1	3	3	2	1			
permanent/strong shelter				1			1		1	3		2	2		2	2	1	1				4	1	2	2	1	1
identification of/access to safe site				1						1		1				4		3	3	1	2				3		2
emergency shelter /tents/tarpaulin				2			1		1	1		1	2		1	1		1	1		1						
building/strengthening flood barrier/canals				1						1		1				2		1							1		1
strengthen/construct road/bridges							2		1	1		1							2		1						
food																			2	1	1	1		1			
rescue assistance boat/lifejackets																2	1	1									
mosquito nets				3																							
cash																			1		1						
early warning system																									1		1
electricity																			1		1						
health facilities																			2								
prepositioning																1			1								
safe shelter for livestock																			1		1						
toilets/hygiene promotion																						1		1			
fast assistance																			1								
insurance for farmers																			1								

TERAI DISTRICTS	BARDIYA			DANG			MAHOTTARI			SUNSARI		
	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY
PREVIOUS EVENTS												
flooding	1	1	1	1	1	1	1	1	1	1	1	1
landslide												
MONSOON NEEDS												
identification of/access to safe site	2		2	1		1	2		2	4	3	1
building/strengthening flood barrier/canals/clearing rivers	1	1		6	2	2				3		2
permanent/strong shelter	2	2					3		2	3	1	2
toilets/hygiene promotion	1	1					3	1	2	1		1
rescue assistance boat/lifejackets	3	1	1	1						2	1	
food				1		1	1	1		2	1	1
strengthen/construct road/bridges							3	3		1		1
water							3	2	1			
restoration of land										2		2
construction materials/CGI							1	1		1		1
emergency shelter /tents/tarpaulin	1		1							1		1
early warning system										2	1	
mosquito nets										1		1
safe evacuation route										1	1	
prepositioning							1					

Shelter Cluster Nepal Monsoon preparedness assessment: Preliminary findings

EXPECTED MONSOON ASSISTANCE



EQ PRIORITY DISTRICTS	BHAKT APUR	DHADING			DOLAKHA			GORKHA			NUWAKO T			OKHALDU NGA			RASUWA			SINDHUP ALCHOK		
	ALL/TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY
PREVIOUS EVENTS																						
flooding	1				1	1	1	1	1	1	1	1	1	1		1				1	1	1
landslide		1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
MONSOON ASSISTANCE EXPECTATIONS - TYPE																						
none	3	4	1	2	1		1	3	1	1												
permanent shelter	1							1	1							1						
cash		1	1																			
CGI														1	1							
road strengthening		1		1																		
safe site														1	1							
food																1						
MONSOON ASSISTANCE EXPECTATIONS - FROM																						
from government	2	1	1		3	1	1	3	2	1	2	1		3	1	2	2	2		2		1
from NGO		1	1		3	1	1	1	1		3	1	1				1		1	5	2	2
promised assistance has not been delivered		2	2		2	1	1	1		1							1		1	1		
from neighbours					1	1																

TERAI DISTRICTS	BARDIYA			DANG			MAHOTTARI			SUNSARI		
	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY
PREVIOUS EVENTS												
flooding	1	1	1	1	1	1	1	1	1	1	1	1
landslide												
MONSOON ASSISTANCE EXPECTATIONS - TYPE												
none	2	1	1	3	1	1	3	2		5	2	2
rescue assistance	1											
MONSOON ASSISTANCE EXPECTATIONS - FROM												
promised assistance not materialised	3	2	1				1	1		2		2
from NGO							2	2				
from government	1						1	1				
assistance has created dependency							1					

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PREVIOUS COPING STRATEGIES USED



EQ PRIORITY DISTRICTS	BHAKT APUR	DHADING			DOLAKH A			GORKHA			NUWAKO T			OKHALD UNGA			RASUWA			SINDHUP ALCHOK			
	ALL/TY	AL L	PT	TY	AL L	PT	TY	AL L	PT	TY	AL L	PT	TY	AL L	PT	TY	AL L	PT	TY	AL L	PT	TY	
PREVIOUS EVENTS																							
flooding	1				1	1	1	1	1	1	1	1	1	1		1				1	1	1	
landslide		1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
COPING STRATEGIES USED IN THE PAST																							
temporary displacement	5	2		1	3	2	1							6	1	4	1	1			3	1	1
rebuilding/clearing roads		2	2		3	2	1	1	1				1		1	1		1	1	2		1	
digging small canals (before and during monsoon)	3				2	1				4	1	2				1		1	3	1	1		
rebuilding schools					2	1	1				1		1	4	1	3							
using locally available/re-using materials		1			2	1					1			1			1			1			
rebuilding water points		3	1	2																			
rebuilding houses					1	1														1	1		
rebuilding canals								1	1														
using alternative markets								1			1												
plastic/straw on the floor	1																						
sharing water sources with other locations		1																					
temporary evacuation														1									
REASONS COPING STRATEGIES NOT USED																							
current site is not safe																	2	1					
lack of money/materials					1									1						1			

TERAI DISTRICTS	BARDIYA			DANG			MAHOTTARI			SUNSARI		
	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY
PREVIOUS EVENTS												
flooding	1	1	1	1	1	1	1	1	1	1	1	1
landslide												
COPING STRATEGIES USED IN THE PAST												
rebuilding/clearing roads	4	2	1	5	2	2	1		1	3		1
temporary evacuation	2	2					5	2	2	5	1	3
temporary displacement	6	2	3	1						2	1	1
rebuilding houses	2	2		2	1	1	2	2				
digging small canals (before and during monsoon)	2		1	1		1						
plastic/straw on the floor	1	1										
REASONS COPING STRATEGIES NOT USED												
lack of money/materials	1	1								1		



Shelter Cluster Nepal Monsoon preparedness assessment: Preliminary findings

CONSTRUCTION MATERIALS AND TOOLS – MARKET ACCESS



Item		UNITS	Bhaktapur -	Dhading -	Dolakha -	Gorkha -	Nuwakot -	Okhaldunga -	Rasuwa -	Sindupalchok -
MARKET			Bhaktapur	Trishuli	Lamidanda	Jaubari	Rautbesi	Siddhicharan	Manigawo	Barabise
				(Nuwakot)		Narayanghat		Dhundh	Ghumti	
						(Chittwan)		Kunda	Trishuli	
						Srinathkot		Kuntadevi	(Nuwakot)	
Bamboo	1 Piece		250	75	75	125	75	200	175	Free
Burnt Bricks	1 Piece		15	14	17	17	26	22	17	17
Cement	1 bag									
	(50kg)		800	975	712.5	712.5	800	1100	850	870
Deformed Steel										
bars - 10-25mm	1 KG	NK		85	82.5	120	75	92.5	84	82.5
welded wire										
mesh (	1 m2		90	200	300	225	NA	400	130	300
Nails for roof	1 KG		127.5	300	150	160	170	280	180	150
Nails for wood	1 KG		95	120	100	110	110	150	225	100
Nylon rope	1 KG		220	150	270	210	190	400	NA	250
Plastic sheet	1 KG		135	305	305	305	210	300	230	350
Tie wire	1 KG		130	150	100	200	160	250	180	100
Timber	1 m3		2800	1400	NA	3500	1400	1000	3000	NA
Claw hammer	1 Piece		250	125	125	125	270	500	240	350
Combination										
pliers	1 Piece		250	225	225	225	275	150	130	325
Gall (Crow bar)	1 Piece	NK		700	700	700	700	1000	3000	1150
Hand saw	1 Piece		150	150	300	120	140	250	350	300
Machete	1 Piece		250	1250	1250	2750	1250	1000	2000	NA
Pick	1 Piece		300	250	330	350	250	300	425	330
Shovel	1 Piece		300	400	350	300	270	300	350	280
Woven sack	1 Piece		7.5	15	25	40	15	50	15	27.5

Item		UNITS	Bardiya - NPR average	Dang - NPR average	Mahottari - NPR Average	Sunsari - NPR average
MARKET			Gulariya	Ghadawa	Jaleswor	Inaruwa
Bamboo	1 Piece		150	100	200	200
Burnt Bricks	1 Piece		12.5	11	13	20
Cement	1 bag (50kg)		800	712.5	700	800
Deformed Steel bars - 10-25mm	1 KG		850	73	78	80
Welded wire mesh standard	1 m2		80	90	80	200
Nails for roof	1 KG		150	160	140	120
Nails for wood	1 KG		150	180	100	100
Nylon rope	1 KG		350	130	250	160
Plastic sheet	1 KG		800	110	260	270
Tie wire	1 KG		150	110	150	120
Timber	1 m3		2400	2300	4500	6000
Claw hammer	1 Piece		150	350	300	250
Combination pliers	1 Piece		250	225	130	250
Gall (Crow bar)	1 Piece		250	1200	200	500
Hand saw	1 Piece		300	180	150	150
Machete	1 Piece				3000	1500
Pick	1 Piece		500	600	350	400
Shovel	1 Piece		650	1350	300	275
Woven sack	1 Piece			10	20	

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PROTECTION



“People in Pokhara south of Semjong village started to buy land in Semjong village to move away from their risk area, cracks were appearing in the land already then. They are now in the process of moving.

People in Ward 9 tried to do the same, to buy land in Pancho village but people in Dharawi village cultivated the land there and did not allow them to move.” [KI LA/SC in Semjong, Dhading, describing the shortage of safe land]



EQ PRIORITY DISTRICTS	BHAKTAPUR	DHADING			DOLAKHA			GORKHA			NUWAKOT			OKHALDUNGA			RASUWA			SINDHULCHOK		
	ALL/TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY
PREVIOUS EVENTS																						
flooding	1				1	1	1	1	1	1	1	1	1	1		1				1	1	1
landslide		1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
LAND TENURE																						
free land	2	1	1		1	1								1		1	4	2	2	1		1
renting land		1		1	1		1	5	2	2	1		1	1		1				2	1	1
own land		3	2	1	3	1	2	1		1	4	2	2	2	1	1				2	1	1
LAND TENURE DOCUMENTATION																						
land documentation - yes		1		1	3	1	2	2	1	1	4	2	2	3	1	2	1	1		2	1	1
land documentation - no	3	3	1	2	1	1		3	2	1	1		1	1		1	4	2	2	2	1	1
risk of eviction/land dispute	2				1	1		4	2	2				1		1	4	2	2	3	1	2
CITIZENSHIP																						
citizenship card - yes	5	4	2	2	4	2	2	4	2	2	4	2	2	3	1	2	4	2	2	4	2	2
citizenship card - no														2		2						
CASTE																						
caste more vulnerable (Janjati)	6	4	2	2	1		1	4	2	2	4	2	2	1	1		4	2	2	4	2	2
caste (Brahmin/Chettri)					1		1	1	1		1	1										
caste (Mixed)					2	2																
caste more vulnerable (Dalit)		1		1																		
INTRA-COMMUNITY TENSION																						
conflict over resources/assistance		1									2		2									

TERAI DISTRICTS				BARDIYA			DANG			MAHOTTARI			SUNSARI		
				ALL	PT	TY	ALL	PT	TY	ALL	PT	TY	ALL	PT	TY
PREVIOUS EVENTS															
flooding				1	1	1	1	1	1	1	1	1	1	1	1
landslide															
LAND TENURE															
free land				4	2	2	2		2	3	1	2			
renting land							1	1		1		1	3	1	2
own land							2	2		2	2		1	1	
LAND TENURE DOCUMENTATION															
land documentation - yes							2	2		3	1	2	1		1
land documentation - no				3	1	2	2		2	1		1	3	1	2
risk of eviction/land dispute				2	1	1				3	2	1	5	3	2
CITIZENSHIP															
citizenship card - yes				3	1	2	4	2	2	3	2	1	2	2	
citizenship card - no				2	1	1				1	1				
CASTE															
caste more vulnerable (Janjati)				1	1		4	2	2	4	2	2	4	2	2
caste (Brahmin/Chettri)				2		2									
caste (Mixed)				1	1										
caste more vulnerable (Dalit)															
INTRA-COMMUNITY TENSION															
conflict over resources/assistance										1		1	2	1	1

Thank you

*Please forward any **additional data sets** or **major modifications** to the proposed methodologies **by COB 14 June***

*Suggestions for **minor modifications and reports/sources** include in contextualisation for the final report are welcome until **COB 19 June**.*

Thank you!

elisabeth.vikman@reach-initiative.org;
+977 981 32 7674; +962 79925 483

